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EDITED BY

Temesgen Gashaw Tarkegn,
Prairie View A&M University, United States

REVIEWED BY

Michael Rush,
Neptune and Company, Inc., United States
Mengqi Jia,
University of Illinois at Urbana–
Champaign, United States

*CORRESPONDENCE

Mohammad Safeeq
✉ msafeeq@ucmerced.edu

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Comparing crop evapotranspiration estimation methods to evaluate water-saving potential of alternative land uses in the San Joaquin Valley, California

Anshika Kandhway¹, Saswata Nandi², Martha H. Conklin¹ and
Mohammad Safeeq^{1,2*}

¹Department of Civil and Environmental Engineering, University of California, Merced, Merced, CA, United States, ²Sierra Nevada Research Institute, University of California, Merced, Merced, CA, United States

This study explores strategies for water conservation by repurposing agricultural land in California's San Joaquin Valley (SJV) for alternative uses like habitat restoration, groundwater recharge, and solar energy development. Water availability in the region has become increasingly constrained due to climate variability and regulatory restrictions under the Sustainable Groundwater Management Act (SGMA), which mandates groundwater sustainability through reduced extraction and enhanced recharge. Climate change and expansion of water intensive high-value crops are further increasing agricultural water demand and exacerbating water scarcity. One proposed water demand management strategy is land fallowing, but leaving land idle can have significant financial and environmental trade-offs. Repurposing fallowed land for alternative uses can address these trade-offs and deliver benefits beyond water savings. Using the VIC-CropSyst, an agrohydrology model, this study assesses water savings across various land-use and climate change scenarios. Results show annual water savings between 370 and 3,740 million m³, requiring the fallowing of 838 to 4,360 km² of farmland, with estimated revenue losses ranging from \$0.87 billion to \$4.4 billion. Habitat restoration and groundwater recharge are identified as the most effective strategies for reducing water demand, while solar development offers moderate savings. Perennial crops, particularly almonds, demonstrate the highest water demand reductions, while corn shows the least. The findings underscore the importance of crop-specific and farm-centric strategies to optimize water use. These insights will aid policymakers in developing multi-benefit land repurposing policies to support sustainable water management in the SJV amidst climate challenges.

KEYWORDS

agrohydrology, crop evapotranspiration, multi-benefit land repurposing, SGMA, water conservation

1 Introduction

Agriculture in California's Mediterranean climate is heavily dependent on irrigation. Water availability has become increasingly constrained due to droughts and regulatory restrictions on groundwater usage. Despite advancements in irrigation efficiency, water availability remains a significant concern, exacerbated by a shift from annual to perennial crops, which strains existing water storage and infrastructure (Medellín-Azuara et al., 2022). Climate change is anticipated to further decrease water availability and increase agricultural water demand (Hanak and Lund, 2012). Implementation of the Sustainable Groundwater Management Act (SGMA), enacted in 2014, has mandated water managers and users to explore strategies for reducing agricultural water demand and bringing groundwater to a sustainable level. A key water conservation strategy that is gaining traction is fallowing the irrigated farmlands or repurposing them for alternative uses (Allen et al., 1998). Because taking land out of agricultural production is a significant financial and land-use decision, a clearer understanding of its impact on water conservation under current and future climate is required.

While the reduction in irrigated land will aid in decreasing groundwater overdraft in line with SGMA, it will also lead to both direct and indirect effects on land use conversion, water and energy uses, soil and water pollution, natural habitats, farming, and disadvantaged communities (Espinoza, Booth et al., 2023; Fernandez-Bou et al., 2023). Land fallowing offers various environmental benefits, including water conservation, drought impacts alleviation, biodiversity promotion, carbon sequestration, land degradation prevention (Żarczyński et al., 2023), and floods management (Bark, 2021). Widespread fallowing of cropland has been used to tackle droughts in California in multiple instances, such as in 1991, 2012–2016, and 2020–2021 (DWR, 1992; Howitt and Sunding, 2003; Rosevelt et al., 2016; Ayres et al., 2022; Medellín-Azuara et al., 2022). More recent work frames such transitions within a multibenefit land repurposing approach, in which irrigated cropland is strategically converted to less water-intensive uses that also support communities and ecosystems (Penny et al., 2025). This practice can also enhance the agricultural system's resilience to climate change by restoring soil fertility through nutrient recycling, improving soil physical properties, increasing organic matter and microbial diversity, and disrupting pest and disease cycles (Schoeneberger et al., 2017; Żarczyński et al., 2023). However, it can indirectly lead to economic and environmental costs, such as rising consumer prices, job losses in the agricultural sector (Hanak, 2003; Howitt and Sunding, 2003), food insecurity (Li et al., 2024), revenue reduction (Dixon et al., 1993), air quality deterioration (Ayres et al., 2022), and potential soil degradation (Scudiero et al., 2016; Corwin, 2021). Additionally, the type of crops grown determines the value of cropland considered for fallowing and the economic losses associated with them (Howitt and Sunding, 2003). Achieving desirable outcomes from agricultural land fallowing while managing trade-offs requires strategic planning and assessment of the water-saving benefits.

Multi-benefit land repurposing in fallowed agricultural lands presents a valuable strategy for achieving ecological, economic, and social gains. Given that land is a finite and multifunctional resource, it is crucial to optimize its use and potential benefits. Instead of leaving agricultural fields fallow, these lands can be repurposed for alternative uses that not only reduce water demand and replenish groundwater supplies but also offer significant ecological and community benefits

(EDF, 2021). The San Joaquin Valley (SJV), known for having one of the highest concentrations of endangered species in the United States (Williams et al., 1998), could significantly improve in species recovery by converting agricultural lands into natural habitats (Stewart et al., 2019). There is an opportunity to expand habitat restoration efforts by integrating them with water conservation objectives. Several studies have explored the potential of converting irrigated farmland with low agricultural productivity into corridors for native species habitat restoration or solar energy generation (Pearce et al., 2016; Butterfield et al., 2013, 2017; Hoffacker et al., 2017; Lortie et al., 2018; McComb et al., 2022). The use of fallowed agricultural land for groundwater banking has been investigated through soil suitability indices (O'Geen et al., 2015; Butterfield et al., 2017). Other alternative uses of the fallowed farmland can include planting rainfed or minimally irrigated crops, grazing, recreation, or parks (Peterson et al., 2022; Quandt et al., 2023). Natural habitats can offer valuable benefits to farmers, such as supporting native pollinators crucial for crop productivity (Kremen et al., 2002; Klein et al., 2012), and fostering natural predators of agricultural pests (Bianchi et al., 2006). As strategic land retirement gains traction, particularly with the implementation of SGMA (Kelsey et al., 2018), the next step should involve support systems for landowners to transition towards these alternative land uses, ensuring that the economic, environmental, and social benefits are fully realized. These benefits become particularly relevant when the water-saving potential of alternative land uses and crop types is quantified. Providing economic compensation for alternative land use encourages water conservation, enhances ecosystem services, and helps protect community health. However, the sustainable implementation of these initiatives faces challenges, including the need for more flexible subsidies, stable participation, better incentives for farmers, provision of off-farm jobs, and addressing the diversity in farmers' opportunity costs (Deng et al., 2021). Incentive programs offered by municipal, state, and federal entities play a crucial role in supporting these efforts. The success of these programs can ultimately be measured by the extent to which they deliver quantifiable benefits such as water savings for the specified land and crop types.

Understanding the water consumption over large areas and diverse land use within irrigated regions will aid in planning water resources and regulating supply. Evaluating the dynamics and variability of crop evapotranspiration (ET_c) offers data-driven support for more efficient land use and water planning across different scales (Wong et al., 2021). Measuring ET_c varies widely in terms of methodology, reliability, resolution, accuracy and cost. There are various methods based on mass or energy balance, soil-water balance, or direct field measurements (Allen et al., 1998). Some of the widely used methods are pan evaporation, lysimeters, eddy covariance, remote sensing, and land-surface models. Depending on the need for study and scalability, one method can be better suited than the other. Ground-based methods for estimating ET_c provide higher accuracy (Levin et al., 2023) but are less suitable for large scale water use assessments due to their high cost, time consumption, and location-specific application (Allen et al., 1998). Unlike ground-based methods, land surface models and remote sensing are less costly, provide extensive spatial coverage, and are useful in large-scale applications. Studies have shown that variations in methodology, accuracy, and spatial resolution can significantly impact agricultural water demand estimations and management decisions (Grantham and Viers, 2014; Espinoza, Bernacchi et al., 2023b). To improve predictions, quantifying uncertainties across different ET_c estimation methods will allow for more

robust, scalable, and reliable approaches to agricultural water planning. Beyond estimating ET_c, process-based agroecosystem and hydrologic models are increasingly used to evaluate how crop management and land-use decisions affect water use, productivity, and the broader water and nutrient cycles. Such models have quantified the trade-offs among crop yield, nutrient losses, and soil carbon across cropping systems (Li et al., 2025) and freshwater availability under combined human and climate pressures at the river-basin scale (Rateb et al., 2026).

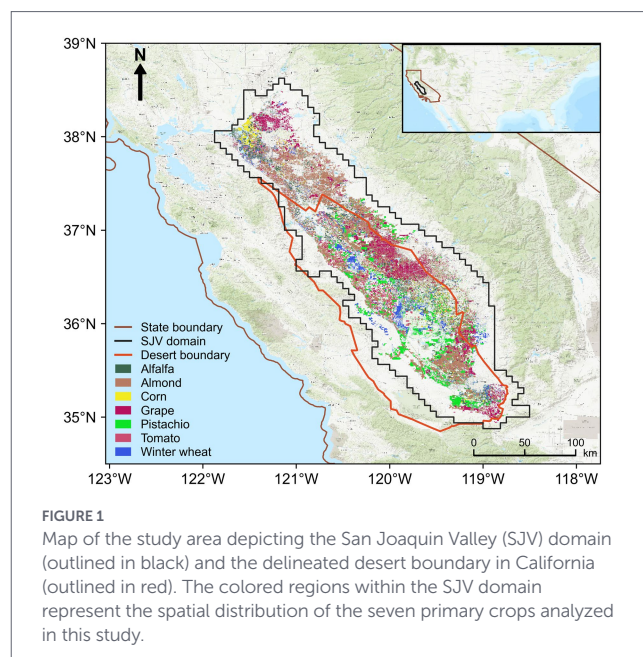
More detailed research is needed to develop an estimation method that accounts for water use across multiple land uses, crop types, and climate scenarios. Studies evaluating the long-term impacts of alternative land use on water conservation in the face of climate change are scarce in California. Since multiple methods exist for evaluating agricultural water demand, their relative advantages and shortcomings should be studied. This study aims to address these critical research gaps through the following objectives: (1) evaluate the uncertainties in ET_c estimation using process-based hydrologic models and remote sensing products, providing a comprehensive understanding of their advantages and limitations in assessing agricultural water demand across various land uses and crop types; (2) quantify potential water savings from different land repurposing strategies, including habitat restoration, groundwater recharge, and solar energy development, under various climate change scenarios, offering insights into the effectiveness of each strategy in conserving water while considering the ecological and socioeconomic implications; and (3) investigate the water savings potential and associated revenue loss of following major crop varieties, providing a comprehensive assessment of the trade-offs between water savings and economic outcomes. By addressing these objectives, this study will aid in developing an assessment plan that identifies long term water conservation under changing climate and land uses when taken for retirement to maximize water savings, ecological benefits, and socioeconomic advantages for stakeholders and communities at large.

2 Methods

2.1 Study area

The SJV forms the southern part of California's Central Valley, extending from the Delta in the north to the Tehachapi Mountains in the south (Figure 1). It is a flat valley floor bordered by the Coast Ranges to the west and the Sierra Nevada to the east, allowing for large-scale mechanized farming and efficient irrigation systems. The region experiences a Mediterranean climate with hot, dry summers and mild, wet winters, with rainfall mainly occurring between November and March and decreasing from north to south. The valley floor, covered in fertile alluvium deposited by rivers over millennia, features a gentle slope ideal for farming.

Land use in the SJV is dominated by agriculture, with urban areas and natural vegetation like forests and wetlands making up a smaller portion. The SJV study domain covers about 43,269 km². Of this, around 23,100 km² (2.31 million hectares) is currently used for irrigated agriculture (USDA-NASS, 2020). The seven crops studied here are almond, grape, pistachio, winter wheat (hereafter wheat), alfalfa, tomato, and corn, selected as the major crops grown in the valley. Together they make up about 65% of the irrigated area in the SJV



(15,246 km² in the 2020 CDL) and account for the majority of its agricultural water use. VIC-CropSyst, the coupled agrohydrology model used in this study (Section 2.2), simulates only those grid cells where one or more of the seven crops cover at least 1% of the cell area, so cells with very small total crop coverage are excluded (see §2.3). This filter removes about 16% of the seven-crop area, leaving a simulated area of 12,759 km² (Figure 1). Per-crop simulated areas are given in Table 1. The water-saving, land-requirement, and economic estimates reported here apply to these seven crops, but caution is warranted in extending them to the full irrigated landscape, which includes other crops that were not simulated.

The San Joaquin desert region in the south-west of the SJV (Figure 1) is the driest part of the valley, receiving 279 mm or less of rainfall annually, with arid soils and extreme temperatures (Stewart et al., 2019). Climate conditions, including precipitation amount and timing, soil moisture, drought, and air temperature, support extensive alkali sink scrub, saltbush shrublands, ephedra scrublands, grasslands, and desert ecosystems. The region is home to several unique and endangered species (Germano et al., 2011), such as the San Joaquin kit fox (*Vulpes macrotis mutica*), the blunt-nosed leopard lizard (*Gambelia sila*), and the giant kangaroo rat (*Dipodomys ingens*).

2.2 VIC-CropSyst model

This research utilized the coupled agrohydrology model VIC-CropSyst (Malek et al., 2017) to assess agricultural water demand across different crop categories (Figure 2). The VIC model, developed by Liang et al. (1994), is a large-scale, process-based model that simulates water and energy balances. It has been calibrated and applied in numerous studies in the Sierra Nevada region of California, including the current study. The CropSyst model (Stöckle et al., 1994, 2003), on the other hand, simulates the growth and development of individual crops based on their characteristics, irrigation, and planting dates to estimate daily evapotranspiration and biomass. By combining these two models, the integrated VIC-CropSyst model generates simulations of soil water balance, crop growth stages, water usage, and yield at a 1/16th degree grid scale (~6 km x ~6 km) and

TABLE 1 Average current crop evapotranspiration, area, volumetric demand and current cropland revenue (\$/m²) under baseline scenario (top 4 rows) and projected crop evapotranspiration (m), reduced area (million m²), reduced demand (million m³) and reduced revenue (\$ million) under four land use scenarios for average and individual crops under the RCP 8.5 climate change scenario.

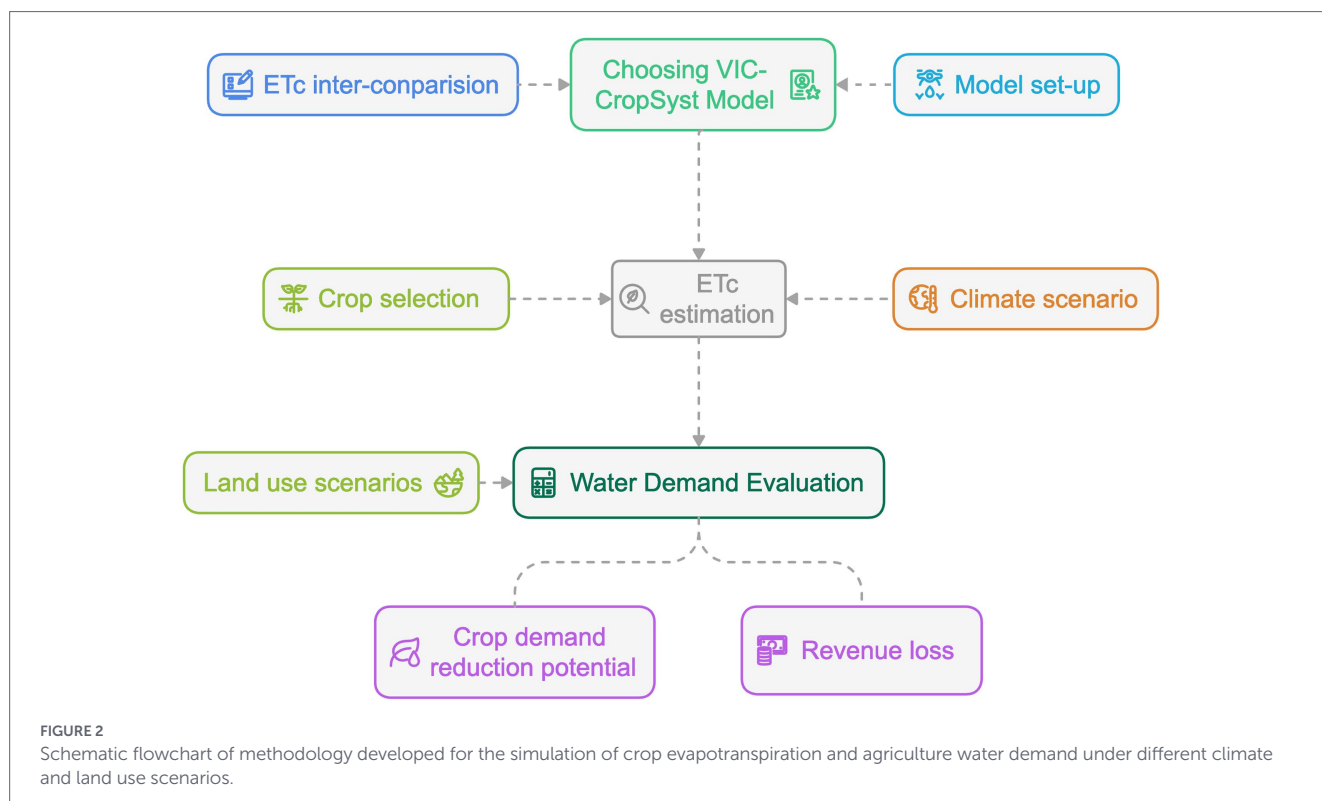
Parameter	Scenario	Average or sum for seven selected crops	Corn	Wheat	Tomato	Alfalfa	Almond	Pistachio	Grape
Current ETc (m)	Baseline	0.9	0.6	0.3	0.6	1.2	1.5	1.2	0.9
Current area (million m ²)	Baseline	12,759*	429	1,299	635	1,319	5,164	2,007	1,906
Current demand (million m ³)	Baseline	14,012*	260	513	369	1,428	7,087	2,499	1,856
Current revenue (\$/m ²)	Baseline	1.026	0.475 [#]	0.202	1.829	0.410	0.710	1.594	1.966
Projected ETc (m)	HBF	0.8	0.6	0.3	0.6	0.9	1.2	0.9	0.9
	HBNF	0.6	0.3	0.3	0.3	0.6	0.9	0.9	0.6
	GWRP	0.7	0.6	0.3	0.6	0.9	0.9	0.9	0.6
	SDP	0.8	0.6	0.3	0.6	0.9	1.2	1.2	0.9
Reduced area (million m ²)	HBF	1,416*	8	125	69	40	579	388	206
	HBNF	4,363*	162	328	295	579	1,700	672	627
	GWRP	3,387*	89	174	129	243	1,582	530	639
	SDP	838*	8	182	134	57	214	178	65
Reduced demand (million m ³)	HBF	1,402*	5	46	36	43	704	388	180
	HBNF	3,137*	63	97	92	351	1,566	559	408
	GWRP	2,799*	43	59	62	216	1,515	491	413
	SDP	729*	5	62	62	60	281	200	59
Reduced revenue (\$ million)	HBF	\$1,606*	\$4	\$25	\$126	\$16	\$411	\$618	\$405
	HBNF	\$4,431*	\$77	\$66	\$540	\$237	\$1,207	\$1,071	\$1,233
	GWRP	\$3,638*	\$42	\$35	\$236	\$100	\$1,124	\$845	\$1,256
	SDP	\$872*	\$4	\$37	\$245	\$23	\$152	\$284	\$128

The water demand is calculated by multiplying crop evapotranspiration with area under each land use. The reduced revenue is the product of current revenue and reduced area under each scenario. Four land use scenarios are: Suitable habitat areas with fallowing history: HBF, suitable habitat areas with no fallowing history: HBNF, suitable habitat sites with good to excellent groundwater recharge potential: GWRP, suitable habitat sites with solar development potential: SDP.

*Indicate sum of seven selected crop; [#]used 2021 data for silage corn as the 2022 and 2023 data were not available.

a daily time step, with the coupling between VIC and CropSyst described in Malek et al. (2017). At each time step, VIC iteratively partitions the available energy among the surface energy-balance components before passing the energy available for transpiration to CropSyst. The coupled model has been evaluated against eddy-covariance flux-tower measurements in irrigated and rainfed fields (root mean square error of 0.8–1.0 mm/day; Malek et al., 2017). A daily time step is appropriate for the seasonal water-demand accounting considered in this study (Allen et al., 1998), and none of the four ETc products compared here resolve sub-daily variation (DWR, 2022; Senay et al., 2013). Phenological stages in this integrated model are determined by accumulated daily heat units in growing degree days, using a linear thermal response curve model, and chilling hours are calculated with the Utah Chilling Unit Model (Richardson et al., 1974). This approach has been successfully used to study the effects of climate change on agriculture in various regions, including California, Washington, and the Columbia River basin (Malek et al., 2017, 2018; Rajagopalan et al., 2018; Yourek et al., 2023; Kandhway et al., 2025).

Model calibration and validation involved identifying sensitive parameters for the VIC-CropSyst model, based on previous studies (Rajagopalan et al., 2018; Scarpore et al., 2022; Adam et al., 2022; and Kandhway et al., 2025). Planting and harvesting dates for annual crops were set, while emergence and dormancy dates were set for perennial alfalfa and tree crops. Seasonal growth stages (flowering, peak leaf area index (LAI), yield filling, maturity, and senescence) for each crop were defined, and heat growing degree-day values were adjusted accordingly. For alfalfa, the stages included leaf emergence, peak LAI, and dormancy. Peak LAI triggered clipping events (assuming 99% removal of aboveground biomass), with dormancy marking the end of the growth cycle. Since alfalfa ETc in the model is derived from modeled reference evapotranspiration, or ETo, and involves multiple clippings, the total seasonal ETc was underestimated compared to observed ETc, although the number of seasonal cuttings and daily ETc followed the observed trends. This underestimation means that the water savings attributed to fallowing alfalfa are, if anything, conservative. Parameters for ETc were specified, including crop coefficient, canopy coverage, leaf water potential, and root depth. Yield-related parameter values,



such as dry yield and maximum fruit load, were also updated. The simulated ET_c was validated against eddy-covariance flux-tower measurements for almond and pistachio at sites in the SJV, with Kling-Gupta efficiency and coefficient of determination close to 0.8 for both crops (Kandhway et al., 2025).

2.3 Input data

The VIC-CropSyst model incorporates various input data, such as climate, land cover, atmospheric CO₂ levels, soil characteristics, crop/vegetation phenology, physiology, and management details. The model was calibrated and validated using gridded climate data, GridMET, from 1980 to 2020 (Abatzoglou, 2013). For future climate projections, downscaled climate data from the Multivariate Adaptive Constructed Analogs (MACA, Abatzoglou and Brown, 2012) were employed, derived from four Global Circulation Models (GCMs) of the Coupled Model Intercomparison Project 5 (CMIP5). These GCMs span a spectrum of climate behaviors: HadGEM2-ES365 (warm/dry), CNRM-CM5 (cool/wet), CanESM2 (average), and MIROC5 (representing variability across all 10 CMIP5 GCMs recommended in California's Fourth Climate Change Assessment). For each of the two Representative Concentration Pathways considered (RCP 4.5, a moderate-emissions scenario, and RCP 8.5, a high-emissions scenario), the four GCMs were averaged to produce a single ensemble per RCP. Atmospheric CO₂ is held constant at 360 ppm (Stöckle et al., 2018) in the modeling so that the differences across GCMs and time periods reflect changes in temperature and precipitation only. CropSyst represents the effect of atmospheric CO₂ on stomatal conductance, and therefore on crop transpiration. Holding CO₂ constant allows the simulated evapotranspiration rate (ET_c) to respond only to warming and changes in precipitation, without the confounding effect of rising CO₂ on the crops (Scarpore et al., 2022). Soil input data were based on STATSGO2 data modified to fit the model's resolution for

textural characteristics, distribution, and hydrology. Initially, soil parameters were calibrated based on Maurer et al. (2002) and later updated using the latest climate averages, with the number of soil layers increasing from 3 to 17 (Kandhway et al., 2025). Land cover information within the SJV boundary was obtained from the USDA-NASS Crop Data Layer (CDL) for the year 2020 at a 30-meter spatial resolution, and the spatial distribution of the seven crops was held fixed at the 2020 CDL throughout the simulation period. Fractional land cover for natural vegetation and crop classes was computed for each 6-km VIC-CropSyst grid cell as the proportion of 30-m CDL pixels of that class within the cell. VIC-CropSyst grid cells were selected for simulation if the seven crops together covered at least 1% of the cell area. Within each selected cell, the crops were simulated as separate tiles in proportion to their fractional area, and the cell-level ET_c was estimated as the area-weighted sum across these tiles. Monthly LAI values for natural vegetation and crops (excluding the chosen 7 crops) were input as fixed values from Bohn and Vivoni (2019) in the vegetation file. Most input and calibration parameters, including approximate phenological stage dates, ET_c, and crop yield information, were gathered from field trials in California, production manuals, published research, reports, and expert consultations.

2.4 ET_c inter-comparison

For analyzing uncertainty in the VIC-CropSyst's simulated ET_c, the outputs were compared with three data-driven products: ET_c derived from California Irrigation Management Information System Spatial CIMIS (hereafter "CIMIS") and two remotely-sensed ET_c products, OpenET SSEBop and VIIRS. The intercomparison was performed with the current-climate average ET_c simulated by VIC-CropSyst with GridMET forcing for the period 1990–2021. The first product, CIMIS, provides a combination of remotely-sensed satellite data with traditional CIMIS station data to produce maps of daily ET_c

on a 2-km grid (DWR, 2022). It uses the Penman-Monteith equation to calculate ETo. The ETc for seven selected crops were calculated by multiplying annual ETo with their average crop coefficient (k_c) values (Sanden, 2010) during the crop's growing season (gs, months) and bare soil coefficient (k_{bs} of 0.15 from Allen et al., 1998) for the non-growing season (ngs, months). The grid-level average annual ETc for the CIMIS was calculated for 2004–2021 period by taking the area weighted average of ETc for crops falling in those grids

$$CIMIS\ ETc = (ETo \times k_c \times gs / 12) + (ETo \times k_{bs} \times ngs / 12) \quad (1)$$

The second product was ETc from OpenET SSEBop (version 0.2.6, hereafter “OpenET”) (Senay et al., 2013; Senay, 2018; Melton et al., 2022). The OpenET implementation uses land surface temperature (Ts) from Landsat (Collection 2 Level-2 Science Products) with key model parameters (cold reference, Tc, and surface psychrometric constant, 1/dT) derived from a combination of climatological average (1980–2017) daily maximum air temperature (Ta, 1-km) from Daymet, and net radiation data from ERA-5 (<https://etdata.org/methodologies/>). The third ETc product was from the USGS available VIIRS dataset (Senay et al., 2013). This dataset's creation involves the integration of VIIRS-based land surface temperature, maximum air temperature from WorldClim, and ETo obtained from gridded weather for global coverage. Both OpenET and VIIRS use the Simplified Surface Energy Balance (SSEBop) model to calculate ETc using remote sensing data. The SSEBop model is a thermal-oriented streamlined surface energy model to calculate ETc using the principles of satellite psychrometry (Senay et al., 2007; Senay, 2018). The monthly raster dataset from OpenET and VIIRS for 2013–2021 period were initially processed on the Google Earth Engine to sum the seasonal water-year ETc.

The average ETc from all four products were downscaled to match the land use data's spatial extent and resolution of 30 × 30 m. Summary statistics, including mean, standard deviation, and interquartile range, were then calculated for each product. Independent sample t-tests were conducted between the VIC-CropSyst model average and each of the other three ETc products. Crop-specific uncertainty analyses were also performed from four sources for the selected seven crops.

2.5 Water demand in alternate land use and climate change scenarios

The baseline scenario consisted of current irrigated acreage (USDA-NASS, 2020) under seven selected crops. Potential land repurposing scenarios were identified based on Butterfield et al. (2017). Habitat corridors for five endangered desert species have been identified within the SJV's desert landscape by Stewart et al. (2019). These areas could potentially provide the scope for habitat restoration and strategic fallowing of irrigated lands in conjunction with other criteria. In all four alternate scenarios, the suitable habitat areas are combined with one of the four additional criteria below. First, among the suitable sites for one or more species, the agricultural areas that were fallowed once or more during the 2011–2015 drought (Melton et al., 2015). These fallowed land parcels during the drought are supposedly the more likely candidates for permanent land retirement (Butterfield et al., 2017). Second, the suitable habitat areas that were not fallowed during the drought could be considered for future fallowing for habitat restoration. Third, suitable habitat sites with good, very good, and

excellent groundwater recharge potential (O'Geen et al., 2015). These zones have been identified using SAGBI which determines suitability for groundwater recharge on agricultural land. The index scores are based on five factors crucial for successful agricultural groundwater banking, including deep percolation, the duration of residence time in the root zone, the land's topography, any chemical constraints, and the condition of the soil surface. Fourth, habitat sites coinciding with least conflict areas for solar energy development (Pearce et al., 2016). These areas include priority, least and potential areas of solar development identified in consensus between environmental, agricultural, and solar energy industry groups. The spatial maps for the baseline and the four alternate land-use scenarios are shown in [Supplementary Figure 2](#) and defined below:

Scenario 1: Baseline (base area with current acreage of crops for water demand).

Scenario 2: HBF (suitable habitat areas with fallowing history),

Scenario 3: HBNF (suitable habitat areas with no fallowing history),

Scenario 4: GWRP (suitable habitat sites with good, very good and excellent groundwater recharge potential),

Scenario 5: SDP (suitable habitat sites having least conflict solar energy development potential).

The spatial layers used to delineate these scenarios are obtained from The Nature Conservancy (Butterfield et al., 2017), the California Soil Resource Lab's SAGBI database (O'Geen et al., 2015), and Pearce et al. (2016), and are combined with the CDL to produce the spatial footprint of each scenario. The average volume of water demand for each raster layer was calculated for the 30 m × 30 m grids and then averaged for the base area of all scenarios, which were kept the same as that of baseline (no fallowing). The four alternate scenarios are applied as static, deterministic changes to the baseline crop distribution, and the temporal component of the analysis comes from the climate forcings. The scenarios are kept static and deterministic so that the resulting differences in water demand can be attributed to land-use change and climate forcing alone; dynamic land-use and economic responses are beyond the scope of this study and are identified as a direction for future work. The VIC-CropSyst modeled ETc for seven select crops from four GCMs (CanESM2, CNRM-CM5, HadGEM2-ES365, MIROC5) was used to average yearly total seasonal water use for the seven crops, and their area weighted average was used to calculate the average water demand under each scenario. The ensemble average ETc and crop acreage data were used to quantify the volume water demand under five alternate land use and two climate scenarios (RCP 4.5 and RCP 8.5). Climate change analysis for crops was expanded to include the gradual changes in the 21st century by dividing into three 30-year intervals (2010–2039 as early, 2040–2069 as mid, and 2070–2099 as late). Then, demand reduction potential of the crops was calculated as the per unit area reduction contributed by each crop by dividing the total volume reduction by each crop in each scenario by the total area under the baseline scenario. Finally, the revenue losses under each cropland fallowing scenario were calculated by multiplying reduced crop area with the per unit value of crop [i.e., total crop value divided by total harvested area for the crop year 2023 (except for corn, which did not have data reported for 2022 and 2023)], expressed in \$/m², acquired from CDFA (2024). In addition, the water-saving cost-effectiveness of each crop and land use scenario was quantified by normalizing the volume of demand reduction by the

TABLE 2 Land-use and climate scenarios evaluated in this study and the outputs reported for each combination.

Land-use scenario	Description	Climate scenarios	Time windows	GCM ensemble
Baseline	Current acreage of the seven crops	Historical, RCP 4.5, RCP 8.5	1990–2021; Early; Mid; Late	4 GCMs averaged
HBF	Suitable habitat with fallowing history	RCP 4.5, RCP 8.5	Early, Mid, Late	4 GCMs averaged
HBNF	Suitable habitat without fallowing history	RCP 4.5, RCP 8.5	Early, Mid, Late	4 GCMs averaged
GWRP	Suitable habitat with good to excellent groundwater recharge potential	RCP 4.5, RCP 8.5	Early, Mid, Late	4 GCMs averaged
SDP	Suitable habitat with least-conflict solar development potential	RCP 4.5, RCP 8.5	Early, Mid, Late	4 GCMs averaged

Scenarios: Baseline (current acreage); HBF (suitable habitat with fallowing history); HBNF (suitable habitat without fallowing history); GWRP (suitable habitat with good to excellent groundwater recharge potential); SDP (suitable habitat with least-conflict solar development potential).

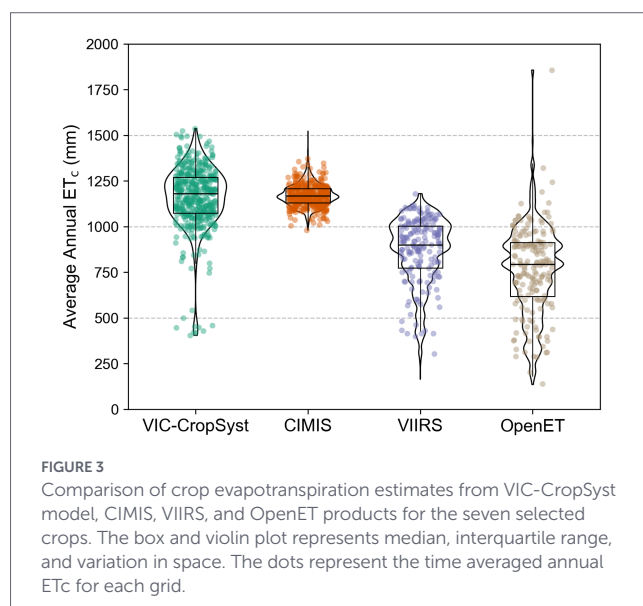
corresponding revenue loss, expressed as water demand reduction per unit revenue loss ($\text{m}^3 \$^{-1}$). Table 2 summarizes the land-use and climate scenarios evaluated and the outputs reported for each combination.

3 Results

3.1 Uncertainty in ETc

The average annual ETc values estimated for the study domain from the four products compared in this study (VIC-CropSyst, Spatial CIMIS, OpenET-SSEBop, and VIIRS-SSEBop) showed distinction from each other. The VIC-CropSyst model's mean ETc (1,160 mm/year) was comparable with the mean value of CIMIS (1,170 mm/year) but was much higher than those of VIIRS (870 mm/year) and OpenET (760 mm/year). The dispersion of data points was observed to be highest in OpenET values and lowest in the CIMIS, suggesting higher spatial variability in the former (Figure 3). The spatial variation (represented by standard deviation) was highest for OpenET (242 mm), followed by VIC-CropSyst (171 mm), VIIRS (182 mm) and lowest by CIMIS (59 mm). The ETc values from VIC-CropSyst, OpenET, and VIIRS exhibited a slight negative skew around the median, while CIMIS values showed a symmetrical distribution. The paired t-test results indicate statistically significant differences between the ETc from the VIC-CropSyst model and the other three products. The t-value was lowest for CIMIS (34), higher for OpenET (194), but highest for VIIRS (870). The *p*-value was less than 0.01 in all instances, so, the difference between the groups is unlikely to be due to randomness. The degrees of freedom were constant in all instances.

The average annual ETc estimated from four sources for the selected seven crops showcased relatively better results from VIC-CropSyst and CIMIS than VIIRS and OpenET (Supplementary Figure 1). The average annual ETc of individual crops had similar median but large variation among the two remote sensing-based products. For instance, the observed ETc from eddy covariance measurement for almonds (1,260 mm/year) and pistachios (1,100 mm/year) (Helalia et al., 2021) were in close agreement with VIC-CropSyst and CIMIS estimates. The close agreement between VIC-CropSyst, CIMIS, and the eddy-covariance measurements was



expected because the model was calibrated using a subset of these products.

3.2 Water demand evaluation

Analysis performed on average ETc, area, and annual water demand reductions under various scenarios revealed varying potential water savings among land use scenarios, but similar results among climate change scenarios (Figure 4 and Table 1). The reduction in acreage under land use scenarios drove the change in reduction in volume of water demand. The total baseline area of 12,759 million m^2 was reduced by 11% in the HBF, 34% in HBNF, 27% in GWRP, and 7% in SDP scenarios. Depending on the land use scenario, in the RCP 4.5 the annual baseline water demand of 14,130 million m^3 can be reduced by 3,159 million m^3 (22%) under the HBNF to the lowest of 737 million m^3 (6%) under the SDP scenario. Similarly, in the RCP 8.5, the base demand of 14,012 million m^3 can be reduced by 3,137 million m^3 (2.54 million acre-feet; 23%) under HBNF to the lowest of 729 million m^3 (0.59 million acre-feet; 6%) under the SDP scenario. Intermediate reductions were estimated for GWRP (20%) and HBF

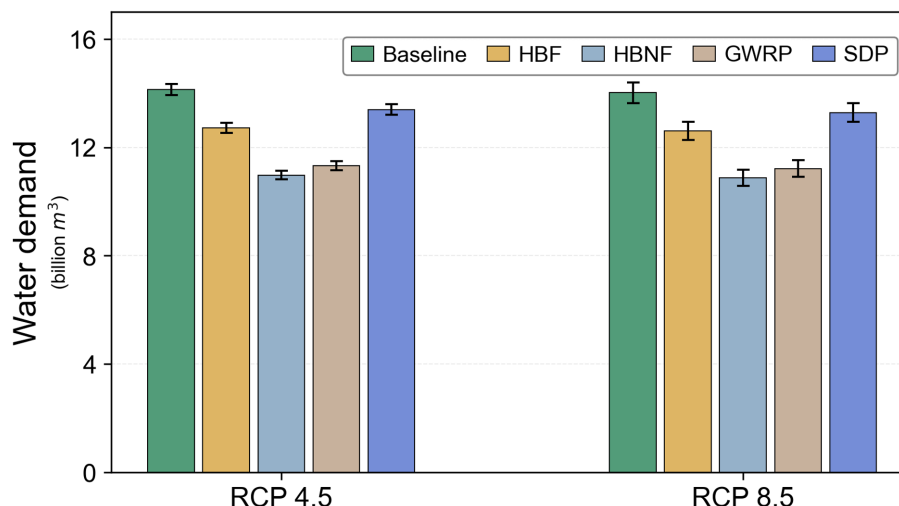


FIGURE 4

Average water demand under five land use (Baseline, HBF, HBNF, GWRP, and SDP) and two climate change (RCP4.5 and RCP8.5) scenarios. Water demand volume under each scenario is calculated by taking ensemble mean crop evapotranspiration simulated by VIC-CropSyst using four global circulation model (CanESM2, CNRM-CM5, HadGEM2-ES365, MIROC5) and spatially weighted by area coverage. Black error bars represent the variation (95% confidence interval) among four GCMs. The four land use scenarios are: Suitable habitat areas with following history, i.e., HBF, suitable habitat areas with no following history, i.e., HBNF, suitable habitat sites with good to excellent groundwater recharge potential, i.e., GWRP, suitable habitat sites with solar development potential, i.e., SDP.

(10%) under both climate scenarios. RCP 4.5 scenarios showed slightly higher reductions in water demands across crops and land use scenarios compared to RCP 8.5. Considering variations among four GCMs and both climate scenarios for each alternate land use scenario revealed deviations of 160 to 350 million m³. Since the values were found to be close for both climate scenarios, further analyses are presented for the extreme-warming scenario, RCP 8.5.

3.3 Crop-specific water demand reductions

Crop-specific water demand reductions, determined by crop ET_c and acreage, were highest for perennial crops, particularly almonds (Figure 5 and Table 1). Baseline water demand was highest for almonds at 7,090 million m³ (5,160 million m² acreage) and lowest for corn at 260 million m³ (429 million m² acreage), reflecting the range of initial water requirements for these crops. Almond showed the highest reduction under HBNF at 1,570 million m³ (22% of base demand by excluding 1,700 million m² of acreage), and lowest under SDP at 280 million m³ (reduced acreage of 210 million m²). Pistachios, grapes, alfalfa, wheat, and tomato followed in order of reduction under HBF, HBNF and GWRP scenarios. In the HBF scenario, pistachios had the second highest demand reduction at 16% from baseline. HBNF scenario yielded significant reductions of 24–25% for corn, alfalfa, and tomato from their baseline values. GWRP scenario showed moderately high reductions for all crops, ranging from 12% for wheat (highest among annual crops) to over 20% for almond, pistachio, and grape crops. SDP scenario resulted in the least reduction for most crops, except for a notable 17% reduction in tomato.

The demand reduction across early, mid, and late periods of the 21st century under climate change scenarios showed changes were influenced by the changing growth cycles and ET_c of the crops in the future climate scenarios. In general, annual crops and almonds exhibited decreasing trends from early to late periods, while perennial crops, including alfalfa, pistachios and grapes, showed slightly increasing trends (Supplementary Figure 3). Demand reductions

decreased more rapidly under wheat across periods compared to corn and tomato crops, while for perennial crops, the rate of change was not significant among all crops and scenarios. These trends reflect both the rising atmospheric demand (ET_o) under warming and the shifts in crop growth cycles (Supplementary Figure 3).

Demand reduction potential per unit area, which is the water saving by the crop divided by the total baseline acreage of the SJV, was higher for perennial tree crops (Figure 6). Under HBF, reduction potentials were relatively modest across all crops, with tree crops demonstrating higher potential than alfalfa and annual crops. HBNF and GWRP scenarios resulted in higher values for the perennial tree crops and alfalfa, with moderate reductions for annual crops. Almonds consistently demonstrated notable potential across all scenarios with reduction potential up to ~0.12 m (m³/m²), while corn exhibited negligible reductions among all scenarios. Among other scenarios, alfalfa had the highest savings under HBNF and GWRP. All crops, including perennial, showed the lowest saving potential under SDP.

Larger reductions in area typically corresponded to higher revenue losses among land use scenarios under extreme climate change (Table 1). HBNF and GWRP scenarios resulted in substantial revenue losses of \$4,431 million and \$3,638 million, respectively. Perennial high-value crops consistently caused significant revenue reductions (\$128 million–\$1,256 million) across all land use scenarios, indicating their economic sensitivity to land repurposing strategies. Grapes, despite having a lower area reduction than almonds, showed higher revenue reduction due to their higher per-unit-area revenue. Alfalfa, with its lower economic value and area coverage, led to moderate revenue reduction compared to other perennial crops. Annual crops demonstrated lower revenue impacts compared to perennial crops. Among annuals, the higher current value for tomato led to higher revenue reductions between \$126 million–\$540 million compared to corn and wheat.

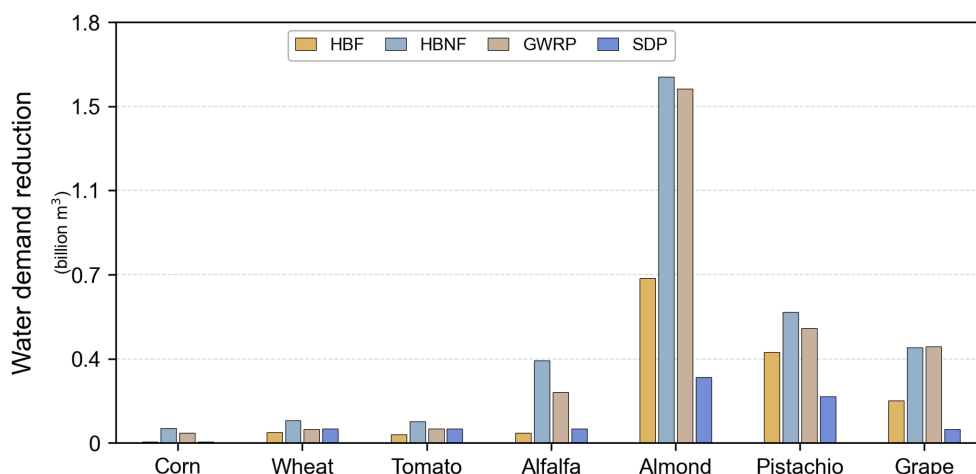


FIGURE 5

Water demand reduction from the baseline water use by each crop for four (Baseline, HBF, HBNF, GWRP, and SDP) land use scenarios under RCP 8.5. Water demand volume for each crop is calculated by taking ensemble mean crop evapotranspiration simulated by VIC-CropSyst with four Global Circulation Model (CanESM2, CNRM-CM5, HadGEM2-ES365, MIROC5) and averaging them spatially for their area coverage. The four land use scenarios are: Suitable habitat areas with following history, i.e., HBF, suitable habitat areas with no following history, i.e., HBNF, suitable habitat sites with good to excellent groundwater recharge potential, i.e., GWRP, suitable habitat sites with solar development potential, i.e., SDP.

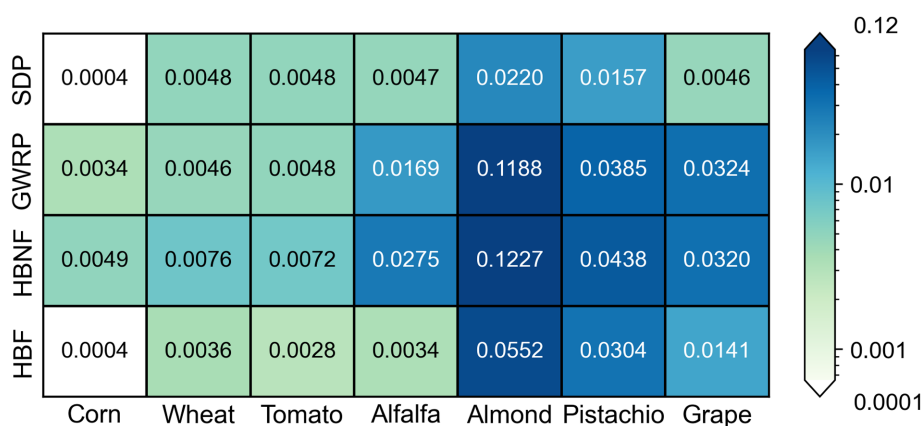


FIGURE 6

Water demand reduction potential (m^3/m^2) for each crop under four (Baseline, HBF, HBNF, GWRP, and SDP) land use scenarios in RCP 8.5. The four land use scenarios are: Suitable habitat areas with following history, i.e., HBF, suitable habitat areas with no following history, i.e., HBNF, suitable habitat sites with good to excellent groundwater recharge potential, i.e., GWRP, suitable habitat sites with solar development potential, i.e., SDP.

3.4 Water demand reduction per unit revenue loss

The water demand reduction per unit revenue loss varied more strongly among crops than among land use scenarios (Figure 7). Alfalfa ($1.5\text{--}2.6 \text{ m}^3 \text{ \$}^{-1}$), wheat ($1.5\text{--}1.8 \text{ m}^3 \text{ \$}^{-1}$) and almond ($1.3\text{--}1.9 \text{ m}^3 \text{ \$}^{-1}$) consistently provided the highest demand reduction per dollar of revenue loss, followed by corn ($0.8\text{--}1.3 \text{ m}^3 \text{ \$}^{-1}$) and pistachio ($0.5\text{--}0.7 \text{ m}^3 \text{ \$}^{-1}$). In contrast, the high per-unit-area value of tomato ($0.2\text{--}0.3 \text{ m}^3 \text{ \$}^{-1}$) and grape ($0.3\text{--}0.5 \text{ m}^3 \text{ \$}^{-1}$) made them the least cost-effective crops for water savings, despite their notable absolute reductions. Almond provided the largest absolute demand reduction while also ranking among the highest reductions per unit revenue loss. At the scenario level, HBF provided the highest water savings per unit revenue loss ($0.87 \text{ m}^3 \text{ \$}^{-1}$), followed by SDP ($0.84 \text{ m}^3 \text{ \$}^{-1}$), GWRP ($0.77 \text{ m}^3 \text{ \$}^{-1}$), and HBNF ($0.71 \text{ m}^3 \text{ \$}^{-1}$). Although HBNF achieved the largest absolute

demand reduction, it retired a greater share of high-value crop-land, lowering its water savings per dollar of revenue loss.

4 Discussion

4.1 Uncertainty in crop evapotranspiration estimation

The comparative analysis of four different methods used for estimating ETC in agricultural landscapes has provided valuable insights into their methodologies, advantages, weaknesses, and resultant variations. The mean ETC from VIC-CropSyst ($1,160 \text{ mm/year}$) and CIMIS ($1,170 \text{ mm/year}$) agreed within 1%, whereas the SSEBop-based VIIRS (870 mm/year) and OpenET (760 mm/year) estimates were 25–35% lower, and these differences can be explained by the

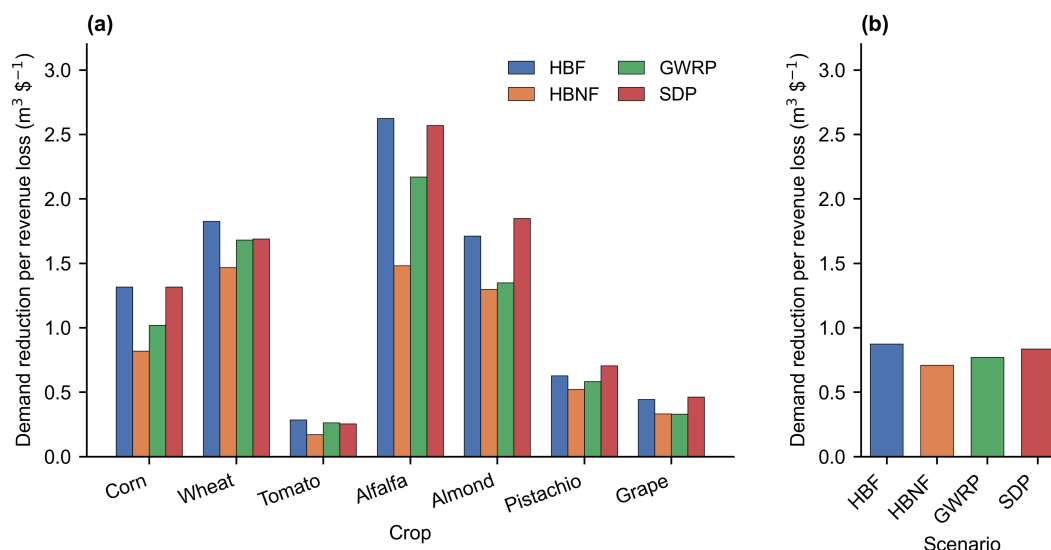


FIGURE 7

Water demand reduction per unit revenue loss ($\text{m}^3 \$^{-1}$) for each crop (a) and for all seven crops combined (b) under four land use scenarios in RCP 8.5. Demand reduction and revenue loss correspond to the reduced demand and reduced revenue reported in Table 1. The four land use scenarios are: suitable habitat areas with following history, i.e., HBF, suitable habitat areas with no following history, i.e., HBNF, suitable habitat sites with good to excellent groundwater recharge potential, i.e., GWRP, suitable habitat sites with solar development potential, i.e., SDP.

input data, modeling techniques and spatial resolutions of the models. CIMIS calculates ETo from ground-based weather stations spread across California and extrapolates the values using remote sensing products. By combining ETo with average crop coefficient values for the selected crops, CIMIS provides a physically based estimate of mean ETc that agrees closely with the VIC-CropSyst model (see Section 3.1) and with eddy-covariance measurements for almond and pistachio (Helalia et al., 2021). While computationally efficient, CIMIS exhibits the lowest spatial variability in output (standard deviation of 59 mm), because a smooth interpolated ETo field is multiplied by a single average crop coefficient per crop (Equation 1), potentially limiting its ability to reflect different soil types and crop irrigation patterns, especially in heterogeneous landscapes. However, physical constraints, parameter uncertainties, and structural errors related to surface resistance measurement and energy imbalance can affect their accuracy (Hu et al., 2021). Moreover, the complexity of the land-surface models requires extensive input data and significant computational resources (Wanniarachchi and Sarukkalgige, 2022), which can limit its widespread application. These models do not place any constraints on crop water use and irrigation happens based on certain thresholds (Malek et al., 2017; Adam et al., 2022). In VIC-CropSyst, this produces a regional mean close to CIMIS but with nearly three times its spatial variability (standard deviation of 171 mm), reflecting simulated differences in soils, weather, and crop growth stages among grid cells. In reality, crops are also subject to water and other stresses, including diseases, pests, and soil salinity, which are not represented in the model. Process-based estimates are therefore likely to overestimate actual ET (Wong et al., 2021), and these stress effects are better captured by remote sensing methods.

OpenET and VIIRS products, using the SSEBop model, provided actual evapotranspiration based on the surface energy balance. These estimates were lower than the average ETc for the region reported in the Cal-SIMETAW unit values (DWR, 2018), possibly due to the combination of multiple products (for climate

and land use) and the absence of irrigation or precipitation data (Kibria et al., 2021). Because the thermal signal integrates the effects of water stress, partial canopy cover, and fallowed fields, the difference of roughly 300–400 mm/year between the SSEBop products and the two demand-based estimates largely reflects the gap between potential crop water demand and actual consumptive use, rather than measurement uncertainty alone. Despite underestimated means, their spatial variability was comparable to or higher than the VIC-CropSyst model (standard deviations of 242 mm for OpenET and 182 mm for VIIRS, compared to 171 mm). Also, the higher spatial resolution of OpenET (1-km) and VIIRS (1-km) compared to CIMIS (2-km) and VIC-CropSyst (6-km) allows for capturing finer details, such as differences in orchard age and canopy density, fallowed parcels, and saline soils, which the coarser grids average out. The SSEBop approach, which omits some components of the surface energy balance, has been reported to perform well with low errors in irrigated fruit and nut orchards in the Central Valley (Knipper et al., 2024; Volk et al., 2024). Data driven, remotely sensed methods for estimating evapotranspiration are consistent and cost-effective for large area applications but require calibration and validation for reliable results across different agricultural profiles (Allen et al., 2007; Hu et al., 2021; Wanniarachchi and Sarukkalgige, 2022). The accuracy of these products depends on the quality and frequency of satellite data. Considering the nature of this study, which requires measuring ETc across a large area with varying climates, soils, growth cycles, and irrigation practices, as well as simulating ETc under future climate and land use scenarios, which remotely sensed products cannot provide, the VIC-CropSyst model's comprehensive approach, low error rates, and ability to account for spatial variability have proven to be valuable for large-scale water resource assessments in heterogeneous agricultural regions. This model effectively balances the need for accuracy with the ability to capture spatial variability, making it well-suited for meaningful applications across distinct land use and crop types.

4.2 Water saving by land use scenarios

Agricultural land repurposing strategies could save approximately 730 to 3,160 million m³ of water annually across the two climate change scenarios, at 0.71 to 0.87 m³ of demand reduction per dollar of revenue loss. Evaluating these strategies revealed significant variations in potential water conservation efficiency under future climate scenarios. Repurposing lands suitable as desert species habitats (HBF and HBNF combined) could save up to 4,540 million m³ of water, but at the cost of losing 45% of irrigated farmland and yielding limited direct economic gain. Fallowing farmland for groundwater recharge saving more than 2,800 million m³ of water, emerged as a particularly impactful approach due to its multiple benefits of reducing water demand, enhancing water availability along with environmental advantages. The estimated total demand for the study domain covering area of 43,269 km² of 54,040 million m³ has the long-term groundwater overdraft of 2,280 million m³ in the SJV (Escriva-Bou, 2019), can be minimized by adopting land fallowing strategy. Depending on land suitability and need, a multifaceted approach that integrates suitable habitat with groundwater recharge or solar energy development can leverage the strengths of each strategy. Such a combined approach can deliver collective economic, environmental, and social benefits.

The results suggested that implementing fallowing on habitat suitable sites and groundwater recharge zones could be most effective in reducing water demand. Large-scale land fallowing under these scenarios consistently showed significant reductions in demand for all crops under both RCP 4.5 and RCP 8.5 scenarios. When normalized by revenue losses, however, fallowing on suitable habitat with prior fallowing history (HBF) delivered the greatest water savings per dollar, indicating that the most cost-effective strategy differs from the one with the largest absolute savings. Solar development sites demonstrated more moderate reductions due to lesser area coverage. The identification of fallowed or retired agricultural land with high habitat suitability potential (Stewart et al., 2019) highlights significant opportunities for desert ecosystem restoration in the SJV. Alongside, multi-functional agricultural landscapes with environmentally friendly practices can enhance the ecological and environmental value of the land while maintaining comparable outputs (Rega et al., 2019), making it vital to examine agricultural production and land-use changes in relation to biodiversity and water conservation. Land-use, land characteristics, and proximity to natural resources govern the environmental outcomes and existence of species in the region (Newman et al., 2017). Studying the systematic association between the occurrence of endangered species and cropland changes at a regional scale can aid in species conservation and incentivization policies.

Integrating groundwater recharge with suitable habitat lands can provide dual benefits, supporting both water sustainability and biodiversity conservation (Stewart et al., 2019). Promoting results from indices that identify potential groundwater zones, such as SAGBI, can be a cost-effective way of helping land and water managers assess recharge viability in their own areas (Bruce et al., 2023). Artificial groundwater recharge is a viable and economically beneficial strategy for ensuring sustained water availability, while mitigating land subsidence (Botzan et al., 1999; Faunt et al., 2016; Ulibarri et al., 2021). Implementing groundwater recharge infrastructure is costly and faces challenges such as land use conflicts, regulatory complexities, and variable effectiveness depending on soil and climate conditions (Botzan et al., 1999; Bachand et al., 2016; Dahlke et al., 2018). An

effective implementation would require evaluating the trade-offs of existing crops on the land (as discussed in detail in the subsequent section), assessing the underlying groundwater quality (Charbonneau and Kondolf, 1993; Waterhouse et al., 2020), and considering the needs of nearby communities (Marwaha et al., 2021). In California, groundwater quality information has been made publicly accessible through the Groundwater Ambient Monitoring and Assessment (GAMA) Program, which centralizes and displays water quality data from various sources on an interactive map (State Water Board, 2023). Another useful tool is the DWR's DAC Mapping Tool (DWR, 2022), which helps locate disadvantaged communities (DACs) across the state. Overlaying the spatial map of suitable groundwater recharge sites with groundwater quality and disadvantaged community data will help identify more suitable areas for recharge.

The overlap between proposed habitat sites and solar energy development areas is minimal, contributing to water savings. The development of utility-scale solar energy is an attractive option for landowners, particularly those with or without water rights, as it provides an opportunity to keep lands after exiting irrigated production (Butterfield et al., 2017). Agrivoltaics, which combines solar power generation with vegetation, has shown potential benefits in enhancing resilience to climate change while providing additional environmental and economic benefits (Choi et al., 2023). Careful site selection can align solar energy development with energy production, food security, and conservation goals (Hernandez et al., 2014, 2015; Owley and Morris, 2019).

Changes in water use from individual land use change scenarios are direct effects of land repurposing, but there are also indirect effects on agricultural inputs and supply, the regional economy, the labor market, and related industries (Sunding and Roland-Holst, 2020; Fernandez-Bou et al., 2023). Water conservation measures should adopt comprehensive methods accounting for both hydrological and financial dimensions (Eslamifar et al., 2024). Additionally, socioeconomic factors like gender, age, education, and income changes significantly influence how land expropriation affects food security (Li et al., 2024). To get a holistic understanding of each land conversion strategy, a literature review was conducted to synthesize and compare their key aspects (Table 3).

Among the three land repurposing strategies for water conservation in the SJV, each approach has its own economic disadvantage. The HBNF and GWRP scenarios resulted in the greatest revenue decline due to the loss of productive, high-value land, indicating that groundwater recharge projects may prioritize low-value land. Habitat restoration enhances biodiversity and supports endangered species but involves high initial costs and potential loss of agricultural revenue. Groundwater recharge effectively saves and augments water supply while providing auxiliary benefits (mitigating drought impacts, improving water quality, and reducing land subsidence), however, requires infrastructure investment and faces regulatory complexities (Cuppari et al., 2024). SDP offers a sustainable approach with minimal revenue loss, supporting water conservation by promoting solar energy on less productive lands. Solar development provides alternative income, renewable energy benefits, dust mitigation, and soil health improvements, though it faces financial challenges and potential land-use conflicts.

Financial incentives and coordinated planning can mitigate its challenges, making it a viable long-term solution for water management in the SJV. Promoting land use change policies will be more effective with incentives and compensation for reducing groundwater

TABLE 3 Overview of land repurposing strategy uses, benefits, drawbacks, challenges, potential solutions, beneficiaries, case studies and references.

Strategy	Habitat suitability	Groundwater recharge	Solar energy development
Use	Habitat restoration	Increase water availability	Source of renewable energy
Benefits	Enhance biodiversity, support endangered species, improves landscape connectivity, supports wildlife conservation, public access to open space and recreation, supports human health and well-being	Enhances water quality, mitigate drought, supports wetland ecosystems, reduces subsidence, improved groundwater quality supports public health for nearby communities	Alternative income, dust mitigation, supports soil health, reduces agricultural risks
Drawbacks	Potential loss of agricultural revenue, initial costs	High infrastructure costs, land use conflicts	High initial costs, potential conflicts with agriculture
Challenges	Financial feasibility, habitat fragmentation, disease spread, introduction of invasive species, permitting and governance complexities	Regulatory complexities, variability in recharge effectiveness, permitting and inter-agency governance	Financial feasibility, regulatory challenges, habitat impact, permitting and governance
Potential solutions	Agri-environmental programs, strategic planning, policies promoting multifunctional landscapes	Financial incentives, continued research and monitoring (using models and geoinformatics), coordinated planning, stakeholder engagement including farmers, policymakers and conservationists	Innovative financing models, policy reforms, agrivoltaics systems
Beneficiaries	Endangered species, ecosystem health	Farmers, water managers, ecosystems	Landowners, renewable energy sector, agriculture

consumption, achieving water savings, and offering ecosystem services. Several success stories and upcoming programs already provide such incentives. In some cases, farmers volunteer and receive compensation through land fallowing agreements to save water or transfer it to other municipalities (Bark, 2021). Several successful programs and initiatives demonstrate the potential of such approaches: (1) Fallowing agreements: Examples include the 1991 Drought Water Bank (DWR, 1992), Forbearance and Fallowing Program Agreement (PVID-MWD, 2004), Imperial Irrigation District Fallowing Program (IID, 2014, 2016), and Colorado River Water Transfer Agreement (SDCWA, 2020); (2) Conservation Reserve Program: This federal program aims to minimize costs, impacts on agricultural production, and enhance ecological benefits by offering compensation for fallowing lands (Miao et al., 2016; Hellerstein, 2017); (3) Multi-benefit Land Repurposing Program (MLRP): Launched in 2022, this program supports the development of less water-intensive agricultural practices and other land uses, creating new job opportunities and stimulating local economies (Department of Conservation, 2021; Espinoza, Bernacchi et al., 2023); (4) LandFlex program: This initiative offers financial incentives to farmers to fallow fields in critically overdrafted groundwater basins and near-at-risk drinking water wells (DWR, 2023b). These programs highlighted the need for accounting groundwater use reduction by land fallowing.

4.3 Crop specific considerations

Water-use reduction varies across crops, with perennial tree crops showing the largest demand reductions and serving as prime targets for water-saving initiatives. The change in demand reduction

by crops was consistent across the early, mid, and late periods of climate change scenarios. The rate at which water saving changes over time is associated with shifts in crop growth cycles as the climate warms. This implies that the long-term effectiveness of water saving by crops will be affected by the changing climatic conditions. The rate of change was smaller in perennial crops, suggesting relatively consistent savings towards the end of the 21st century compared to annual crops. Because atmospheric CO₂ was held constant and crop management and adaptation were not varied (Section 2.3), factors such as elevated-CO₂ effects on transpiration (Scarpore et al., 2022) were not isolated, and the long-term advantage of perennial crops should be interpreted within this scope.

The analysis of water use and reduction per unit area across different crops provided their efficiency under various land repurposing strategies. For the total crop area of 12,759 million m² under seven selected crops, the base water usage per unit area serves as a benchmark for comparison across different reduction strategies. This metric is valuable for understanding the intensity of water savings on a per-unit-area basis, helping to identify which crops and strategies yield the highest savings per unit of land, and providing practical insights for optimizing water use. Land fallowing on suitable habitat with groundwater recharge proves effective for water-intensive crops like almonds and pistachios. Crop selection among the land use strategies also plays a significant role in enhancing recharge and reducing groundwater use. For instance, recharging groundwater by flooding alfalfa fields with highly permeable soils during winter has been proven effective, causing only minor crop and yield damages (Dahlke et al., 2018). Field experiments have shown that almost 90% of applied irrigation can contribute to groundwater recharge. In the Central Valley, the conversion of 34 million m² (8,350 acres) of

agricultural land for solar installation is primarily driven by the low opportunity cost associated with low-value field and hay crops (59%) (Stid et al., 2022).

The negative trade-offs and economic impacts of saving water by fallowing perennial croplands are substantial. Boser et al. (2024) showed that fallowing of 5% of high-water-use agricultural lands can reduce water consumption significantly, comparable to crop switching strategies. However, fallowing lands currently used by these crops or switching to less water-demanding crops like grains or hay poses challenges due to market and operational costs, reduced farm revenues and job losses. Moreover, fallowing of land with labor-intensive crops can affect rural and low-income communities who rely on agricultural jobs, leading to lower household spending and further economic decline. Overall, while fallowing land occupied by perennial crops can be a major step for sustainable water management, it requires careful planning to balance economic, environmental, and community needs.

The indirect impacts of taking hay and field crops out of production have cascading repercussions, affecting downstream industries such as transportation and food processing (Sunding and Roland-Holst, 2020). Alfalfa, corn and wheat crops are used as feed for the dairy sector, which is the most profiting agriculture industry in California. Though these indirect impacts are not accounted for, this can be taken up for future study. Alternative to permanently fallowing some of these crops, a middle path can be making temporary land fallowing as a part of crop rotation. For example, wheat, which can be grown in the SJV with minimal irrigation, can serve as an alternative to land fallowing (Peterson et al., 2022). Similar innovative strategies should be explored to reduce the tradeoffs of idling land for conserving water. Adopting crop-specific, incentive-driven strategies will enable tailored water management, benefiting stakeholders, including farmers, conservationists, and policymakers.

5 Conclusion

This study explored the potential of repurposing agricultural land to enhance ecological benefits and the associated revenue losses. By comparing four approaches for estimating water demand (ETc) in agricultural landscapes, the study highlighted their methodologies, strengths, limitations, and the uncertainty inherited from model inputs and modelling choices. While land fallowing conserves water, it can lead to economic, social, and environmental challenges. Strategic repurposing, however, can provide significant benefits such as biodiversity conservation, ecological connectivity, water sustainability, and renewable energy opportunities. Successful implementation requires balancing trade-offs by evaluating crop value, water use, sector impacts, and infrastructure costs. To optimize water-saving strategies while maintaining economic sustainability, this study recommends the following approaches: (1) prioritize fallowing of water-intensive, low-value annual crops; (2) implement rotational fallowing or crop switching strategies for perennial crops to minimize economic impacts; (3) adopt innovative irrigation techniques and crop varieties to sustain productivity with reduced water use; (4) integrate groundwater recharge projects with agricultural operations during suitable seasons; (5) develop economic models that account for both

direct and indirect impacts of land-use changes on the regional economy; (6) encourage solar energy projects on marginal agricultural lands to provide alternative income streams for farmers; and finally, (7) introduce policies and incentives to support farmers transitioning to water-efficient practices or alternative land uses. Future research should evaluate the long-term impacts of water reduction, particularly the benefits of habitat restoration, groundwater recharge, and aligning solar energy projects with agricultural and environmental objectives for sustainable land-use practices.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

AK: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. SN: Conceptualization, Data curation, Formal analysis, Methodology, Software, Visualization, Writing – review & editing. MC: Supervision, Writing – review & editing. MS: Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/frwa.2026.1879797/full#supplementary-material>

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